Chapter 4: Cells, the Units of Life

(Grade 8 life science course notes, written by Mr. Haddad for his class)

Every living thing is made of one or more cells. A cell is the smallest unit that can carry out all the processes of life: taking in energy, getting rid of wastes, growing and reproducing.

4.1 The cell theory

Scientists in the 1600s and 1800s built the cell theory from many observations with microscopes.
It has three parts:
1. All living things are made of one or more cells.
2. The cell is the basic unit of structure and function in living things.
3. All cells come from cells that already exist.

Robert Hooke named cells in 1665 after looking at thin slices of cork. Anton van Leeuwenhoek was the first to see living single-celled organisms. Later, Schleiden (plants), Schwann (animals) and Virchow (cells come from cells) put the pieces together.

4.2 Prokaryotes and eukaryotes

Prokaryotic cells, such as bacteria, have no nucleus; their DNA floats in the cytoplasm.
Eukaryotic cells, such as plant, animal and fungus cells, keep their DNA inside a nucleus and have other membrane-bound organelles. Prokaryotic cells are usually much smaller than eukaryotic cells.

4.3 Organelles and their jobs

cell membrane: a thin, flexible layer that controls what enters and leaves the cell.
cytoplasm: the gel-like material inside the cell where many chemical reactions happen.
nucleus: holds the cell's DNA and directs the cell's activities.
mitochondria: release energy from food (sugar) in a process called cellular respiration.
ribosomes: build proteins. They are found in every cell, prokaryotic and eukaryotic.
endoplasmic reticulum: a folded network that moves materials around the cell.
Golgi apparatus: packages proteins and sends them where they are needed.
vacuole: stores water, food and wastes. Plant cells usually have one large central vacuole.
lysosomes: break down worn-out cell parts and wastes (mostly in animal cells).

4.4 Plant cells and animal cells

Plant cells have three structures most animal cells do not:
- a cell wall outside the membrane that gives support and shape,
- chloroplasts, where photosynthesis turns light energy into chemical energy stored in sugar,
- a large central vacuole that helps the plant stay firm.
Both kinds of cells have a membrane, cytoplasm, a nucleus, mitochondria and ribosomes.

4.5 Levels of organization

In many-celled organisms, cells are organized in levels:
cells -> tissues -> organs -> organ systems -> organism.
Example: muscle cells form muscle tissue; muscle tissue, with other tissues, forms the heart (an organ); the heart and blood vessels form the circulatory system.

Review questions from class (Mr. Haddad, period 3)

1. Which part of the cell theory did Virchow add?
2. Name two structures found in plant cells but not in animal cells.
3. Why do muscle cells have many mitochondria?
4. Put these in order from smallest to largest: organ, cell, organ system, tissue.

Vocabulary: cell, cell theory, organelle, nucleus, membrane, cytoplasm, mitochondria, chloroplast, cell wall, vacuole, ribosome, prokaryote, eukaryote, tissue, organ, organ system